NEC

NONLINEAR MODEL | UPA845TC

BJT NONLINEAR MODEL PARAMETERS[(1)]

Parameters	Q1	Q2	Parameters	Q1	Q2
IS	3.5e-17	3.7e-17	MJC	0.15	0.18
BF	109.7	81	XCJC	0.1	0.5
NF	0.9896	0.9944	CJS	0	0
VAF	38.8	27.7	VJS	0.1	0.75
IKF	9.55	0.98	MJS	0	0
ISE	1.83e-11	2.53e-10	FC	0.37	0.5
NE	3.8	30	TF	6e-12	5e-12
BR	3.8	12.3	XTF	0.1	0.2
NR	0.9965	1.002	VTF	0.1	0.1
VAR	1.8	2.9	ITF	0.001	0.001
IKR	90	7.29	PTF	0	0
ISC	6e-15	1.35e-15	TR	1e-9	1e-11
NC	1.17	1.215	EG	1.11	1.11
RE	1	1	XTB	0	0
RB	12	8	XTI	3	3
RBM	3.5	4	KF	0	0
IRB	0.007	0.001	AF	1	1
RC	21.4	13.9			
CJE	0.4e-12	0.49e-12			
VJE	0.99	0.83			
MJE	0.23	0.18			
CJC	0.13e-12	0.12e-12			
VJC	0.57	0.6			

(1) Gummel-Poon Model

UNITS

Parameter	Units
time	seconds
capacitance	farads
inductance	henries
resistance	ohms
voltage	volts
current	amps

MODEL RANGE

Frequency: 0.1 to 4.0 GHz

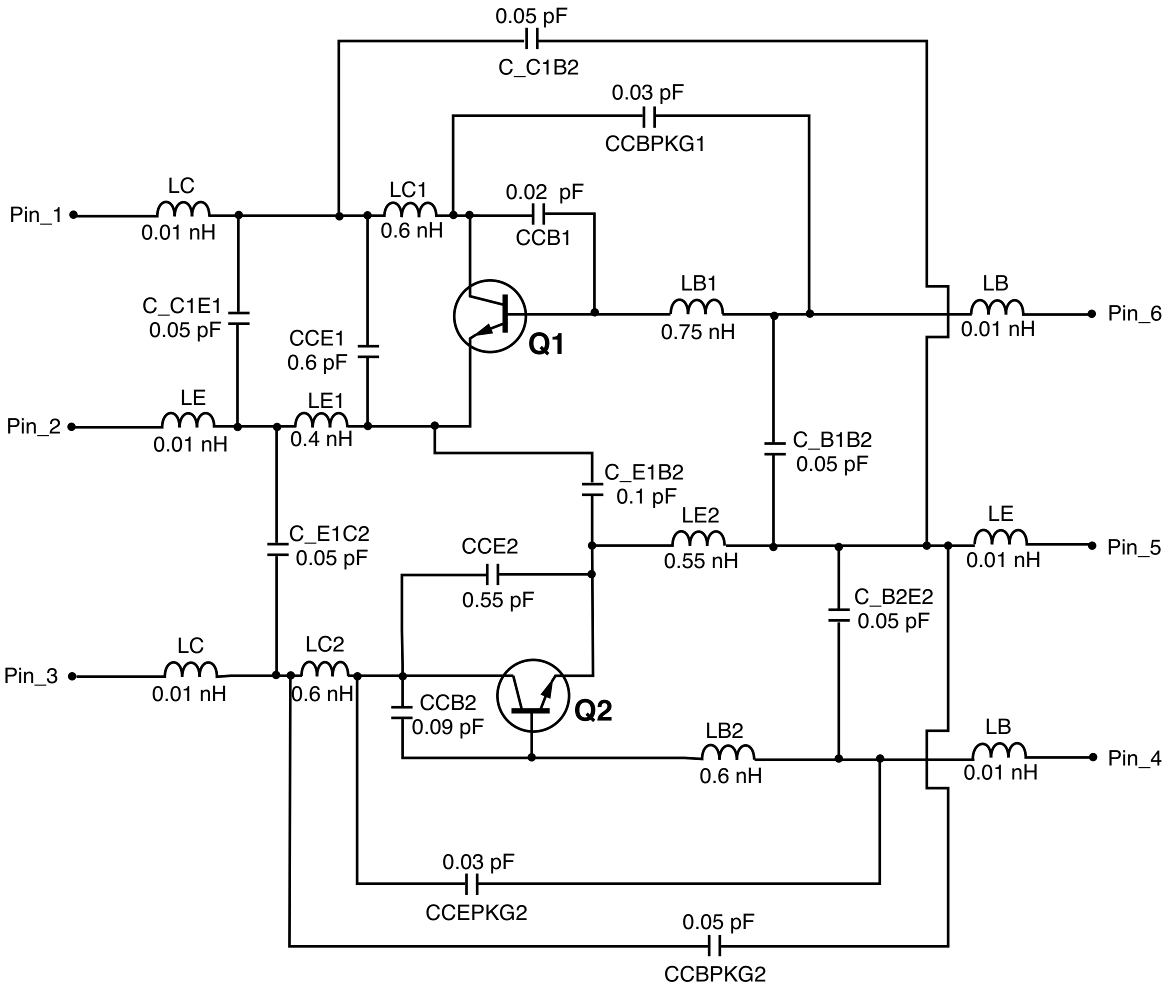
Bias: V_{CE} =0.5 V to 2.5 V, I_C = 1 mA to 20 mA

Date: 04/01

Note:

This nonlinear model utilized the latest data available. See our Design Parameter Library at **www.cel.com** for this data.

SCHEMATIC



MODEL RANGE

Frequency:	0.1 to 4.0 GHz
Bias:	V_{CE} = 0.5 V to 2.5 V, I_C = 1 mA to 20 mA
Date:	04/01